

Q1. Harshad number

ANSWER

```
n = int(input("Enter a number: "))
s = sum(int(i) for i in str(n))
if n % s == 0:
    print("Harshad Number")
else:
    print("Not a Harshad Number")
print("\n Harshad Numbers between 1 and 500:")
for i in range(1, 501):
    if i % sum(int(j) for j in str(i)) == 0:
        print(i, end = " ")
```

Q2. GCD

CODE

```
a = int(input("Enter first: "))
b = int(input("Enter second: "))
while b != 0:
    temp = a
    a = b
    b = temp % b
print("GCD is", a)
```

INPUT

```
24
18
```

OUTPUT

```
Enter first: Enter second: GCD is 6
```

Q3. sum to a target value

ANSWER

```
def find_pairs(1st, target):
    pairs = []
    for i in range(len(1st)):
        for j in range(i + 1, len(1st)):
            if 1st[i] + 1st[j] == target:
                pairs.append((1st[i], 1st[j]))
    return pairs
data = list(map(int, input("Enter numbers: ").split()))
t = int(input("Enter target: "))
print("Pairs:", find_pairs(data, t))
```

Q4. Bitwise Operators

ANSWER

```
a = 0b1101
b = 0o25
c = 0x1a

result = (a & b) | c

print(f"a={a}, b={b}, c={c}, result={result} ")
print(bin(result), oct(result), hex( result))
print(result >> 2, result << 1, ~result)
```

Q5. binary search

CODE

```
def rotate_binary_search(arr, target):
    low = 0
    high = len(arr) - 1

    while low ≤ high:
        mid = (low + high) // 2

        if arr[mid] == target:
            return mid

        if arr[low] ≤ arr[mid]:
            if target ≥ arr[low] and target < arr[mid]:
                high = mid - 1
            else:
                low = mid + 1
        else:
            if target > arr[mid] and target ≤ arr[high]:
                low = mid + 1
            else:
                high = mid - 1

    return -1

data = [4, 5, 6, 7, 0, 1, 2]
t = int(input("Target: "))

result = rotate_binary_search(data, t)

if result ≠ -1:
    print("Found at index:", result)
else:
    print("Target not found")
```

OUTPUT

Target: Found at index: 4